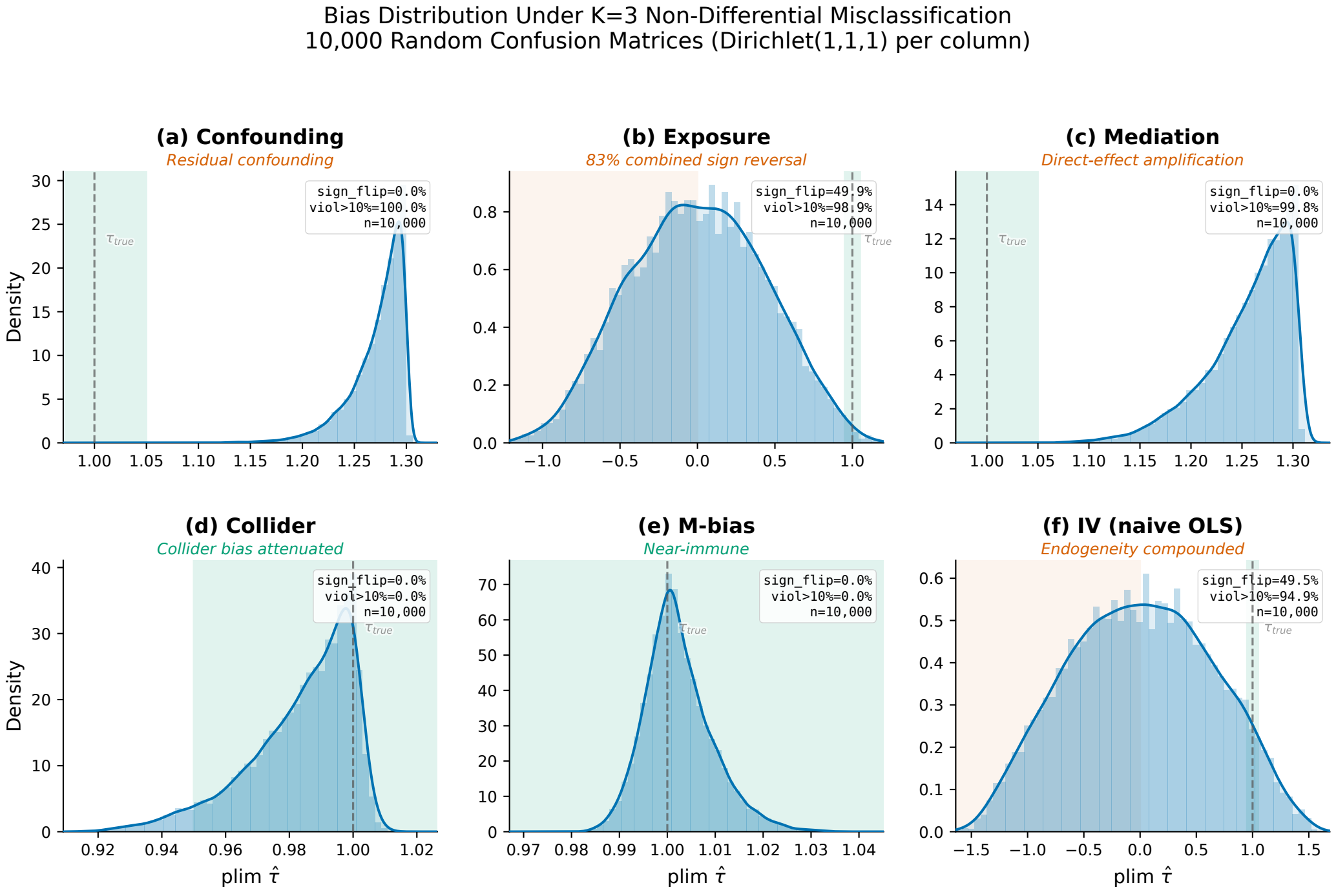

Bias Distribution Under K=3 Non-Differential Misclassification
10,000 Random Confusion Matrices (Dirichlet(1,1,1) per column)
(a) Confounding
Residual confounding
sign_flip=0.0%
viol>10%=100.0%
n=10,000
(b) Exposure
83% combined sign reversal
sign_flip=49.9%
viol>10%=98.9%
n=10,000
(c) Mediation
Direct-effect amplification
sign_flip=0.0%
viol>10%=99.8%
n=10,000
(d) Collider
Collider bias attenuated
sign_flip=0.0%
viol>10%=0.0%
n=10,000
(e) M-bias
Near-immune
sign_flip=0.0%
viol>10%=0.0%
n=10,000
(f) IV (naive OLS)
Endogeneity compounded
sign_flip=49.5%
viol>10%=94.9%
n=10,000
Density
plim $\hat{\tau}$
τ_{true}